

Chapter-06: Database Design using ER Model

CSE-317: System Analysis and Design

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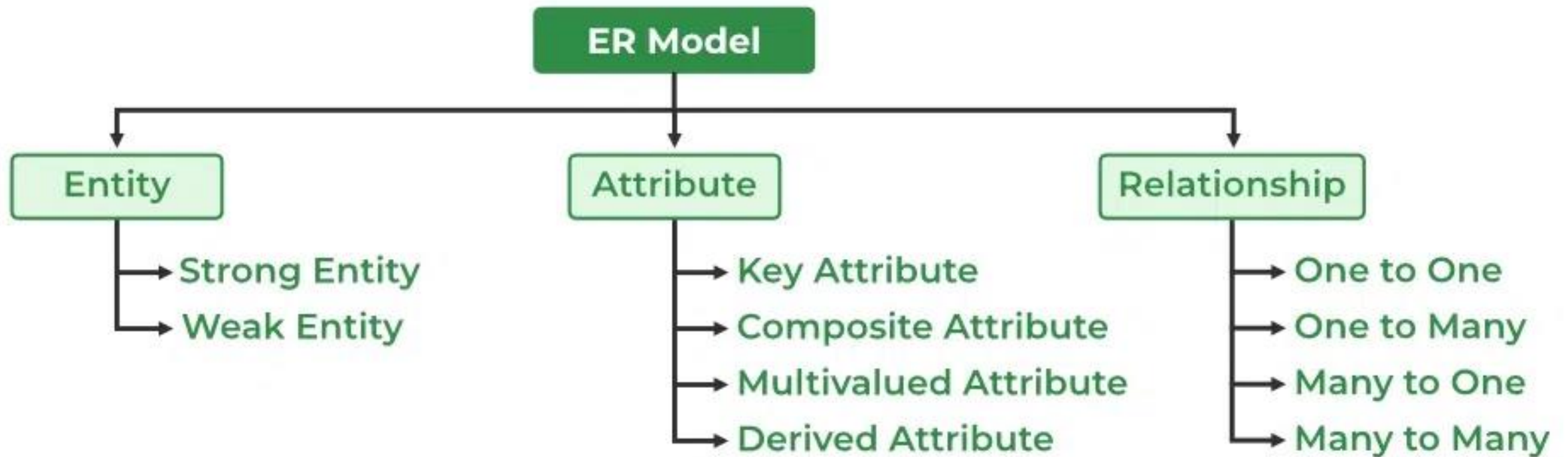
Entity-Relationship Model

- The Entity-Relationship Model (ER Model) is a **conceptual model for designing a database.**
- This model represents the **logical structure of a database,**.

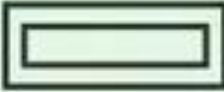
Introduction of ER Model

There are **three parts** in the ER Modeling or Diagram.

- **Entity:** An object that is stored as data.
 - E.g: *Student, Course, or Company.*
- **Attribute:** Properties that describe an entity.
 - E.g: *StudentID, CourseName, or EmployeeEmail.*
- **Relationship:** A connection between entities.
 - E.g: *Student enrolls in a Course.*



Symbols Used in ER Model

Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Entity

- An Entity represents a real-world object.
- **Tables** in relational database represent these entities.
- Example of entities:
 - **Real-World Objects:** Person, Car, Employee etc.
 - **Concepts:** Course, Event, Reservation etc.
 - **Things:** Product, Document, Device etc.

Types of Entity

- There are two main types of entities:
 - Strong
 - Weak

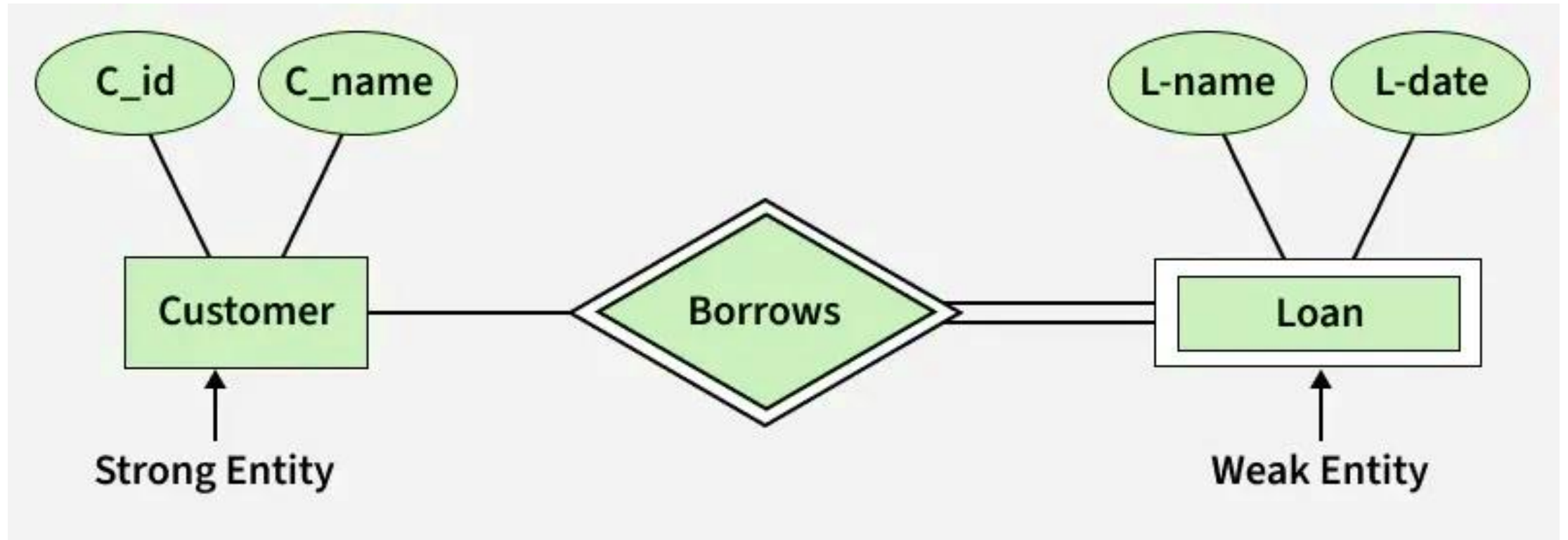
Strong Entity

- It has a **key Attribute** that can uniquely identify.
- Does **not depend** on any **other Entity** for identification.
- It has a **primary key** that ensures its uniqueness.
- It is represented by a **rectangle** in an ER diagram.

Weak Entity

- **Can't be uniquely identified** by its own **attributes** alone.
- **Depends on a strong entity** (identifying entity) to be identified.
- Represented by a **double rectangle**.
- **Participation** is always **total**.
- The relationship is called **identifying relationship**.
- Represented by a **double diamond**.

Example



Attributes

- Attributes are the **properties that define the entity type**.
- For example, for a Student entity Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student.
- In ER diagram, the attribute is represented by **an oval**.



Types of Attributes

- Key
- Composite
- Multivalued
- Derived

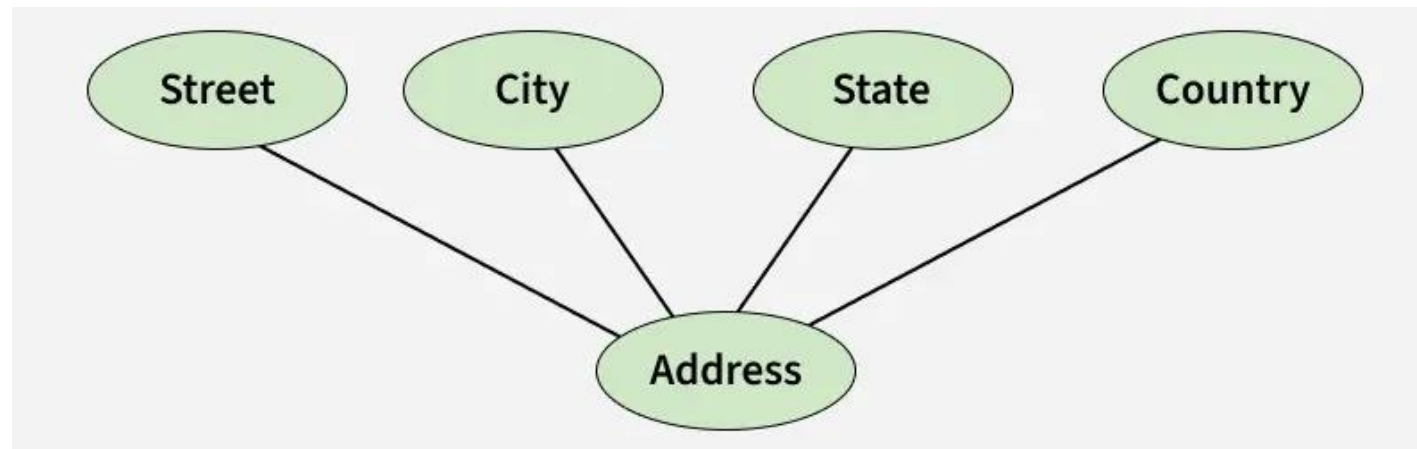
Key

- The attribute which **uniquely identifies each entity** in the entity set is called the key attribute.
- In ER diagram, the key attribute is represented by an **oval with an underline**.



Composite

- An attribute **composed of many other attributes** is called a composite attribute.
- In ER diagram, the composite attribute is represented by an **oval comprising of ovals**.



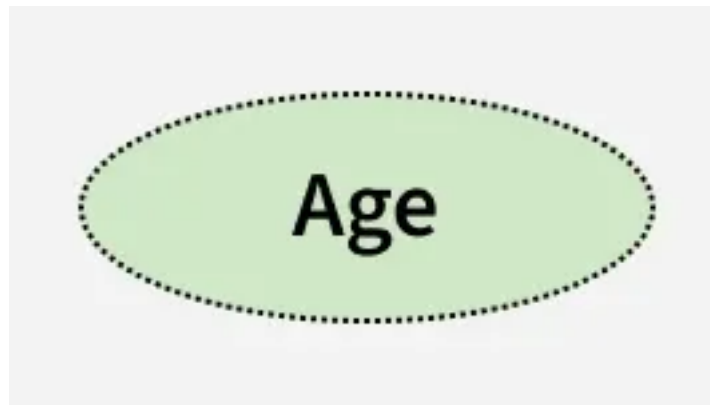
Multivalued

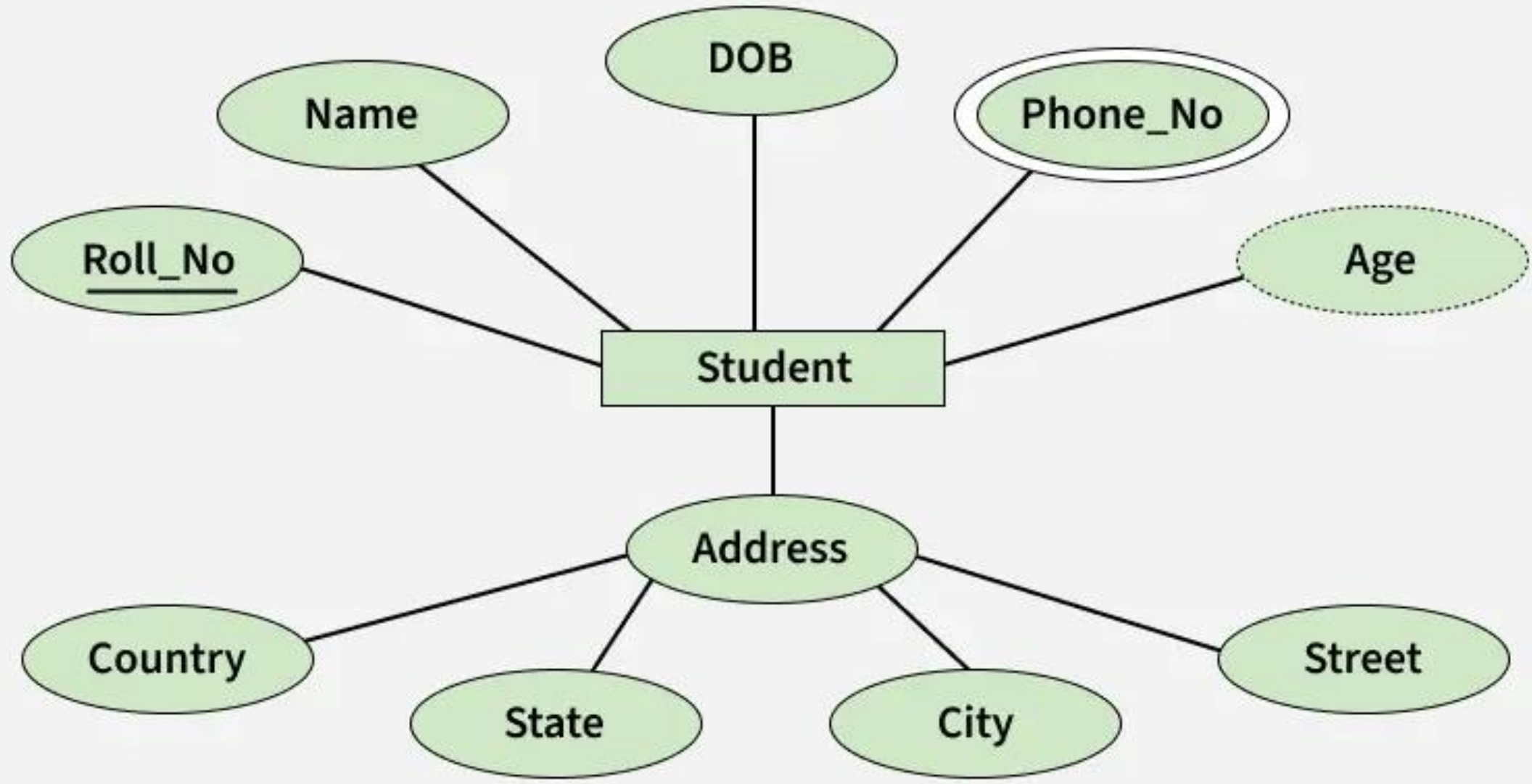
- An attribute **consisting of more than one value** for a given entity.
- For example, Phone_No (can be more than one for a given student).
- In ER diagram, a multivalued attribute is represented by a **double oval**.



Derived

- An attribute that can be **derived from other attributes** of the entity type is known as a derived attribute.
- e.g.; Age (can be derived from DOB).
- In ER diagram, the derived attribute is represented by a **dashed oval**.





Relationship Type and Relationship Set

- A Relationship Type represents the association between entity types.
 - For example, 'Enrolled in' is a relationship type that exists between entity type Student and Course.
- In ER diagram, the relationship type is represented by a **diamond** and connecting the entities with lines.

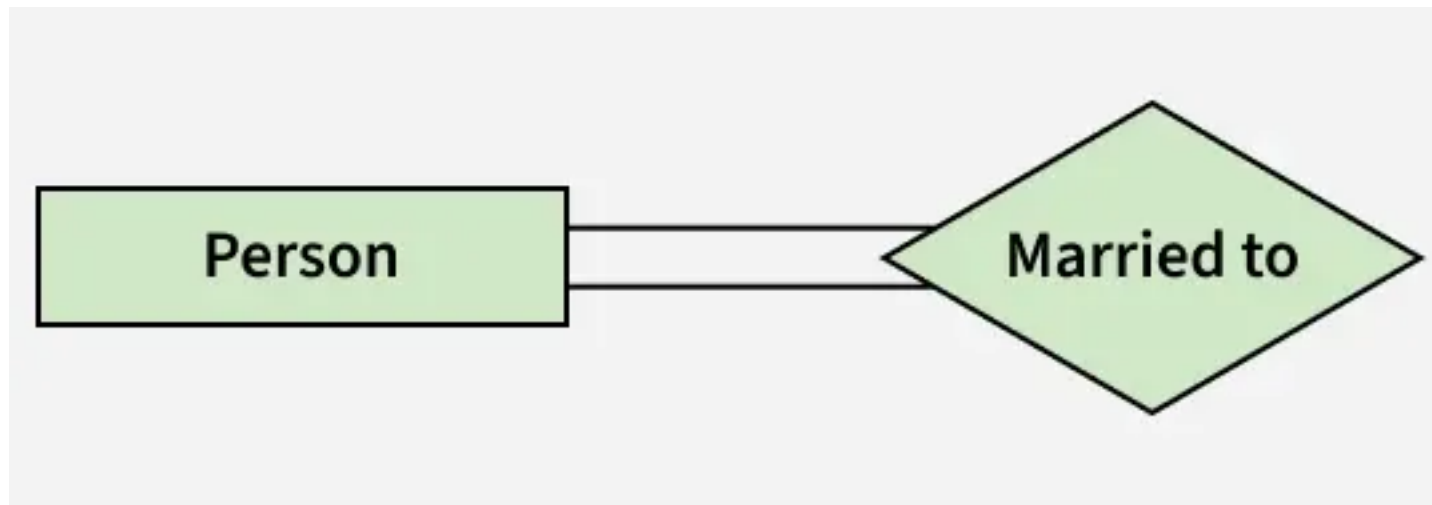


Degree of a Relationship Set

- The number of different entity sets participating in a relationship set is called the degree of a relationship set.

Unary/Recursive Relationship

- When there is only ONE entity set participating in a relation, the relationship is called a unary relationship.
- For example, one person is married to only one person.



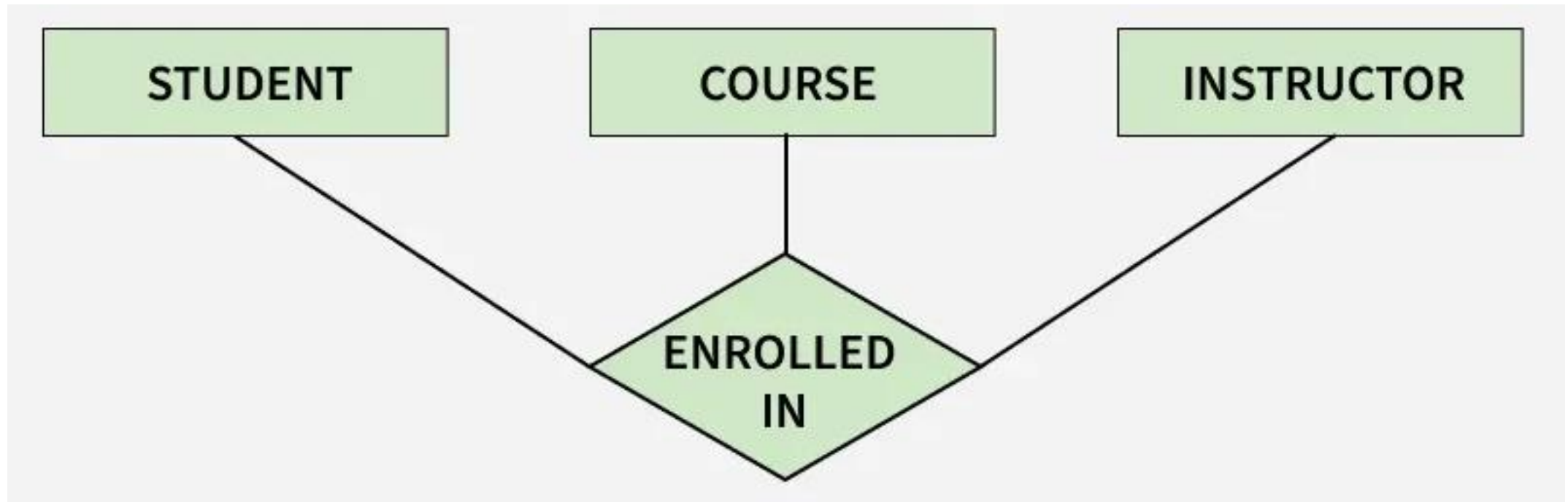
Binary Relationship

- When there are TWO entities set participating in a relationship, the relationship is called a binary relationship.
- For example, a Student is enrolled in a Course.



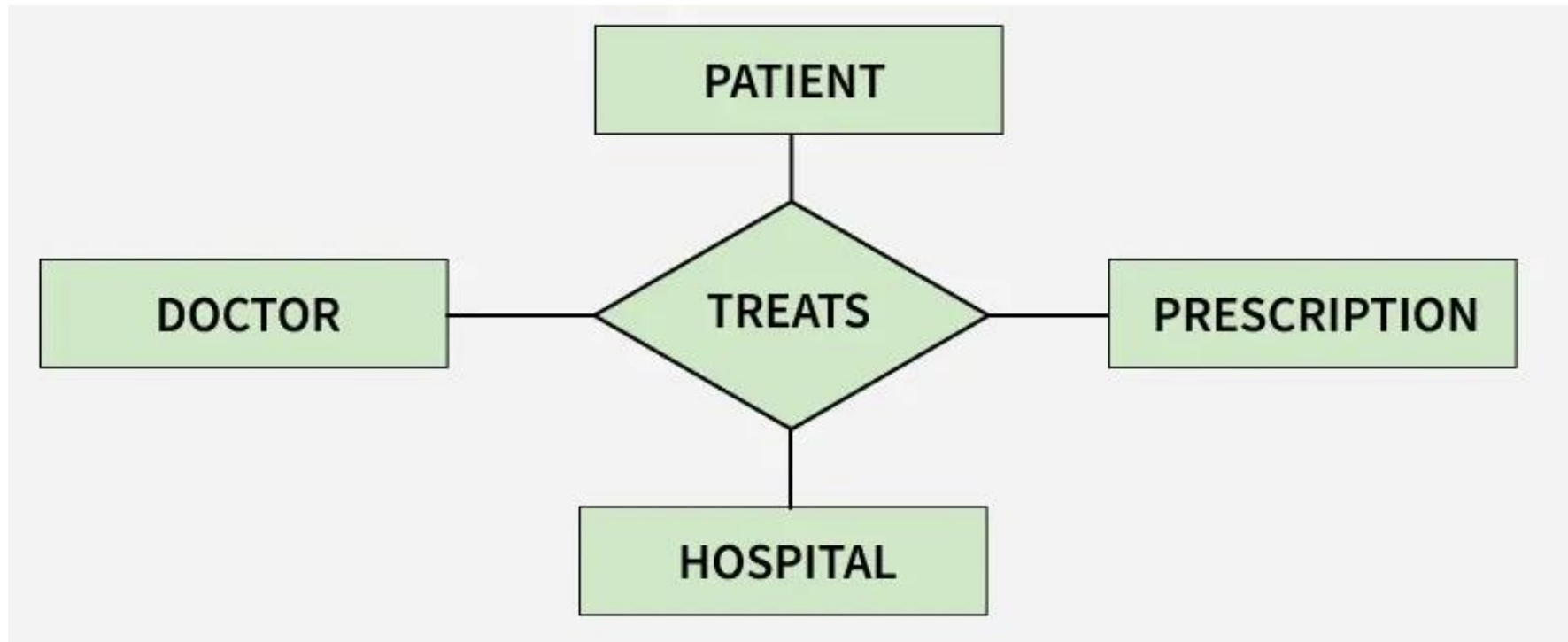
Ternary Relationship

- When there are three entity sets participating in a relationship, the relationship is called a ternary relationship.



N-ary Relationship

- When there are n entities set participating in a relationship, the relationship is called an n-ary relationship.

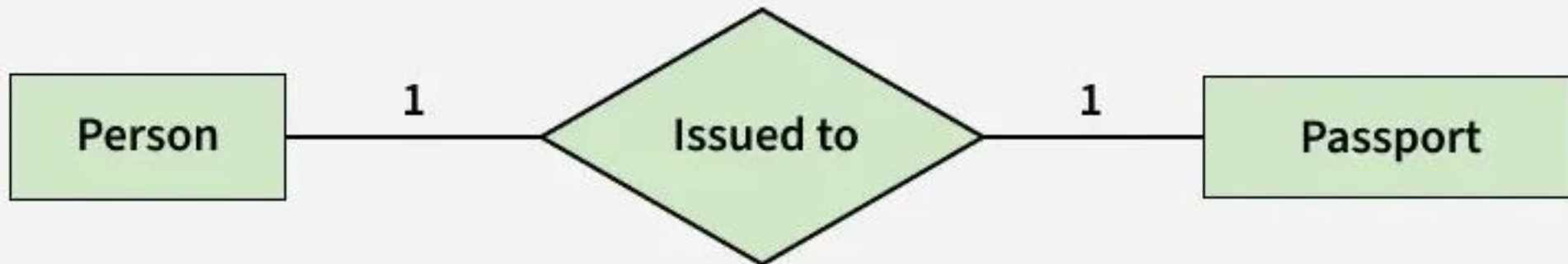


Cardinality in ER Model

- The maximum number of times an entity of an entity set participates in a relationship set is known as cardinality.
- Cardinality can be of different types:

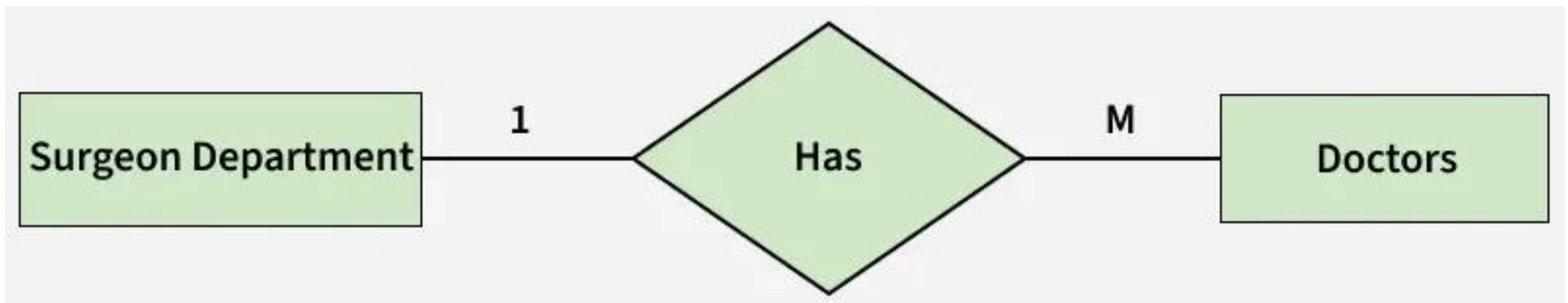
One-to-One

- When each entity in each entity set can take part only once in the relationship, the cardinality is one-to-one.
- Let us assume that one person can be issued only one passport, and one passport is issued to only one person.
- So, the relationship will be One-to-One (1 : 1), meaning that each person has a single passport, and each passport belongs to a single person.



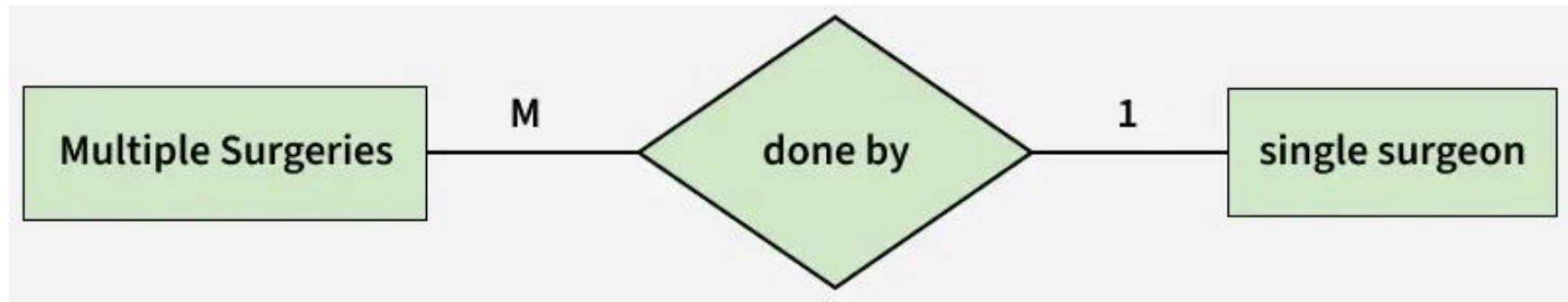
One-to-Many

- In a one-to-many relationship, one entity can be associated with multiple entities.
- For example, a single Surgeon Department can have many Doctors. Therefore, the cardinality of this relationship is 1 to M, meaning one department can have many doctors.



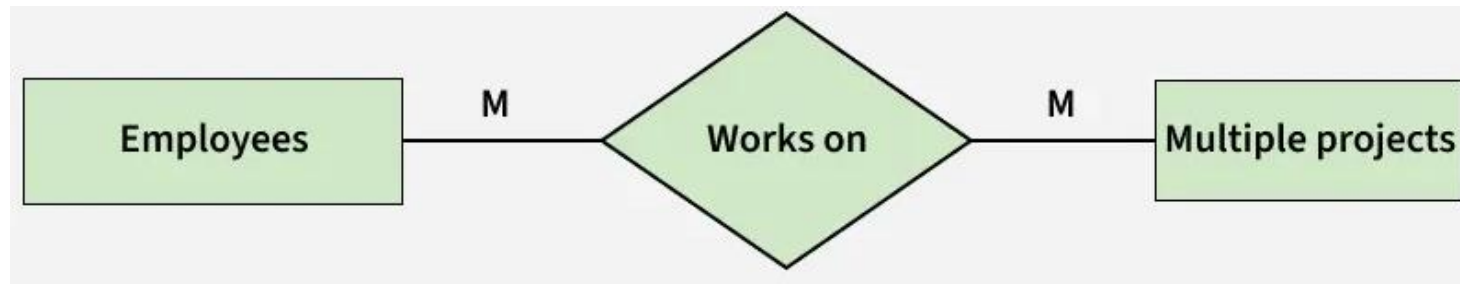
Many-to-One

- When entities in one entity set can take part only once in the relationship set and entities in other entity sets can take part more than once in the relationship set, cardinality is many to one.
- Let us assume that multiple surgeries can be performed by one surgeon, but one surgery is performed by only one surgeon.
- So, the cardinality will be M to 1, meaning that many surgeries can be done by a single surgeon, but each surgery is done by only one surgeon.



Many-to-Many

- When entities in all entity sets can take part more than once in the relationship cardinality is many to many.
- Let us assume that an employee can work on multiple projects and each project can have multiple employees working on it.
- So, the relationship will be many-to-many (M:N), meaning that one employee may be associated with several projects, and one project may involve several employees.

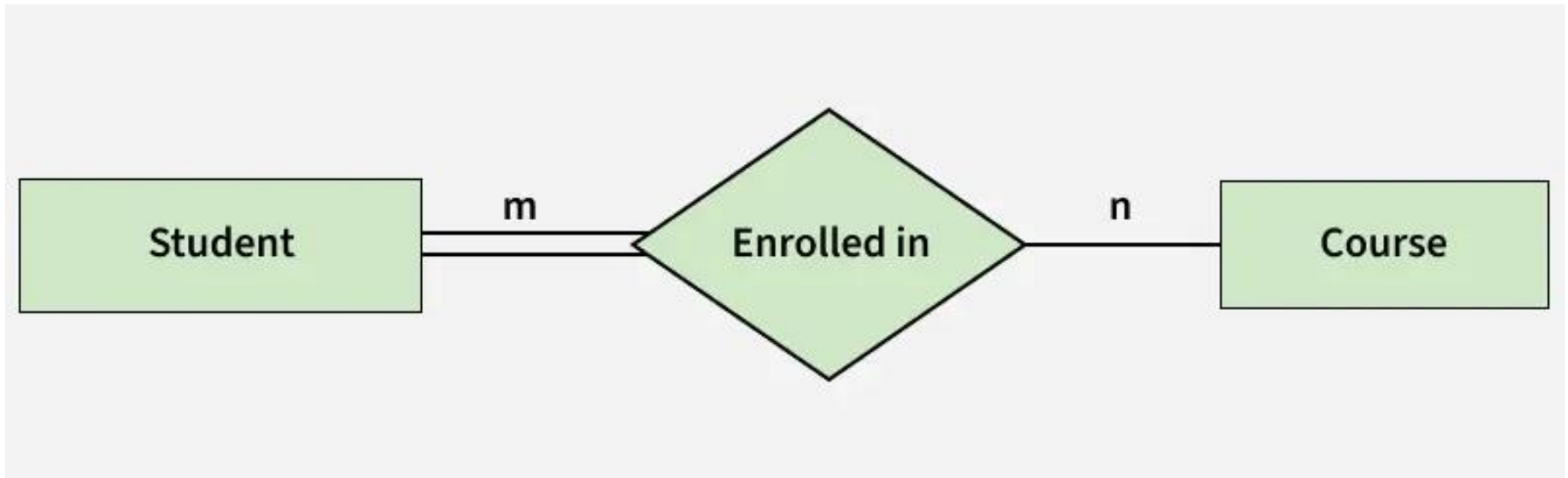


Participation Constraint

- Participation Constraint is applied to the entity participating in the relationship set.
- **Total Participation:**
 - Each entity in the entity set must participate in the relationship.
 - If each student must enroll in a course, the participation of students will be total.
 - Total participation is shown by a double line in the ER diagram.
- **Partial Participation:**
 - The entity in the entity set may or may NOT participate in the relationship.
 - If some courses are not enrolled by any of the students, the participation in the course will be partial.

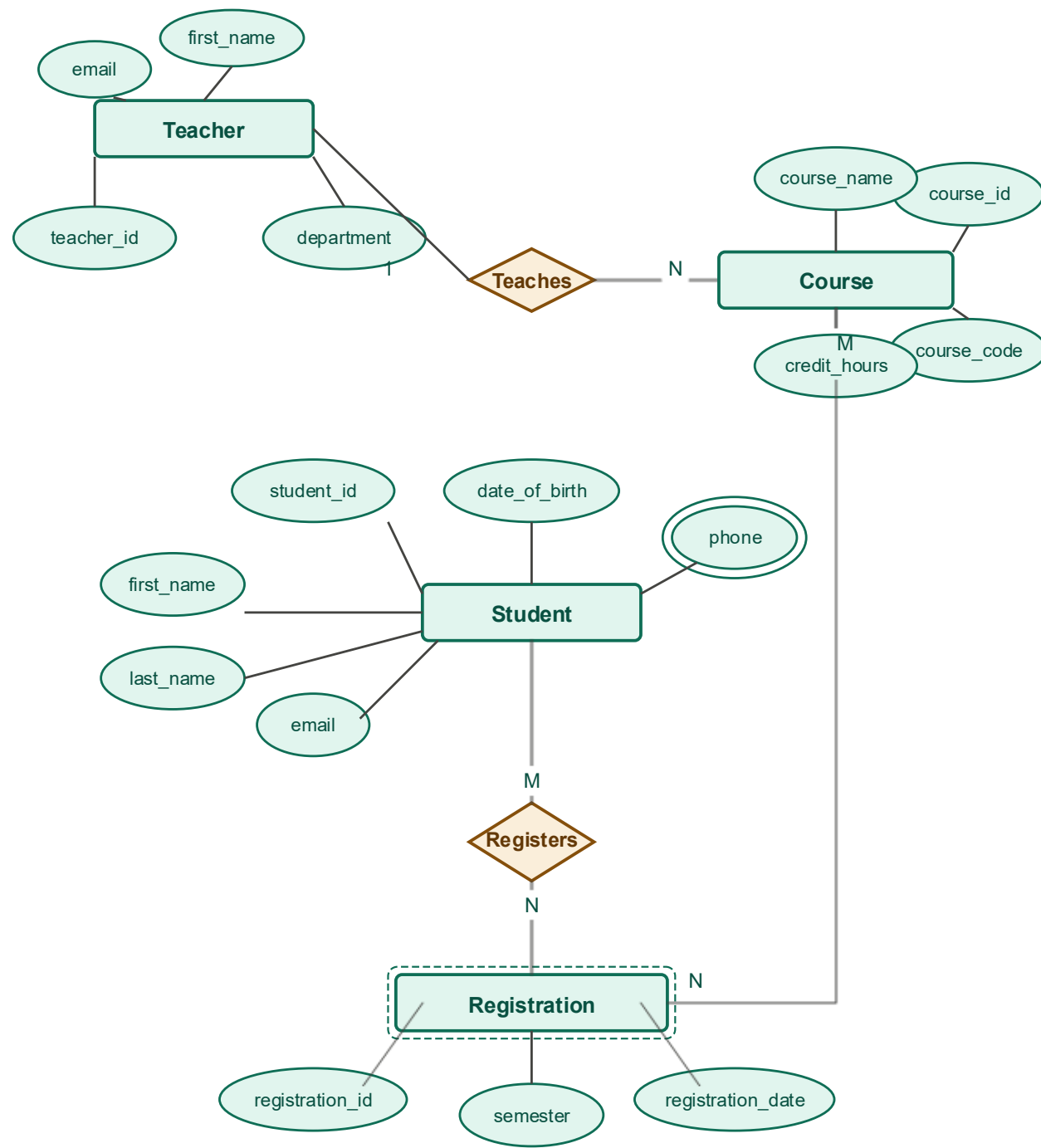
Participation Constraint

- The diagram depicts the 'Enrolled in' relationship set with Student Entity set having total participation and Course Entity set having partial participation.



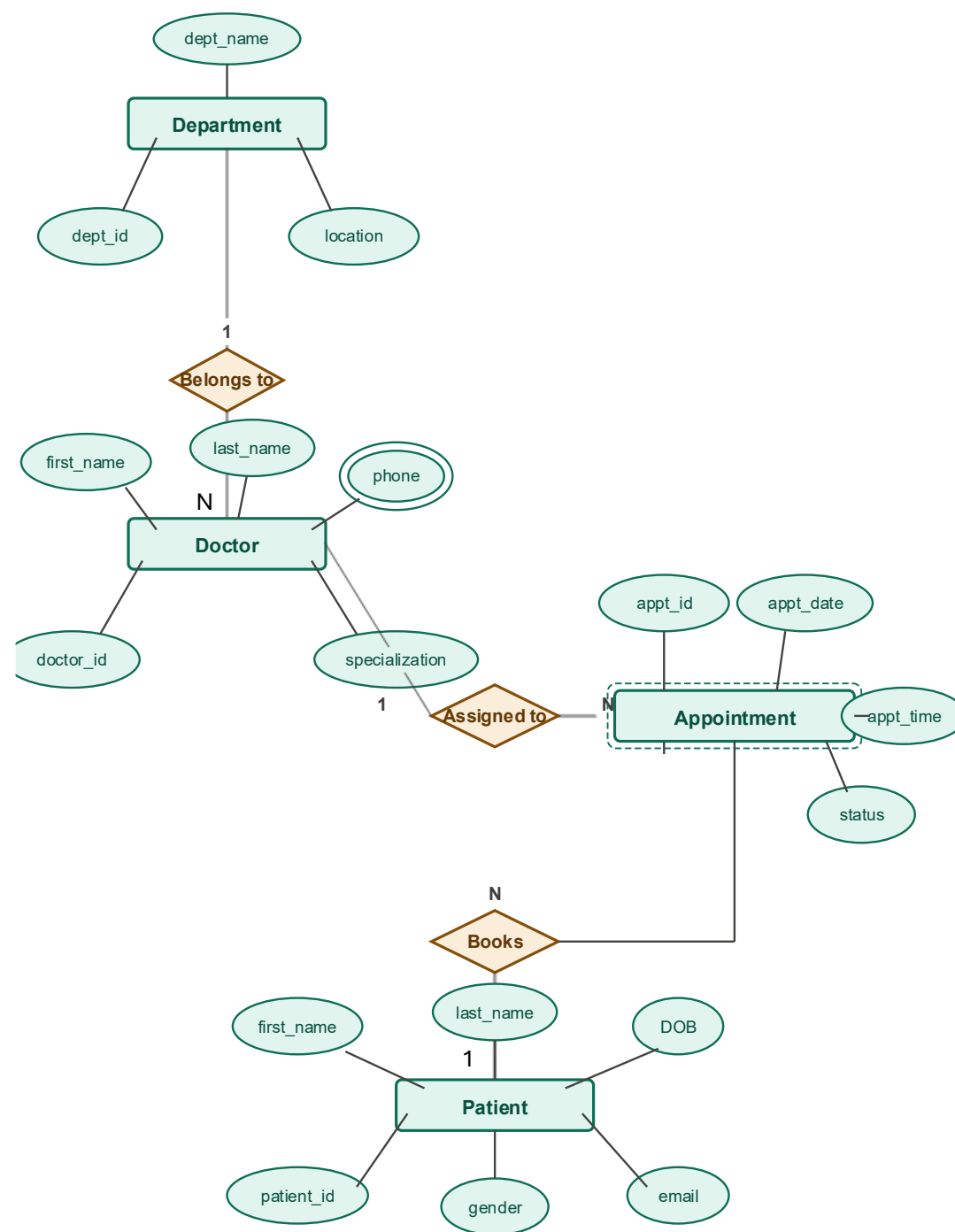
Draw an Entity Relationship Diagram (ERD) according to the following scenario using proper ERD rules, including entities, attributes, primary keys, relationships, and cardinalities.

- A university course registration system stores information about students, courses, teachers, and registrations. A student can register for many courses, and one course can be registered by many students. Each course is assigned to one teacher, but one teacher can take many courses. Each registration stores the semester and registration date.



Draw an Entity Relationship Diagram (ERD) according to the following scenario using proper ERD rules, including entities, attributes, primary keys, relationships, and cardinalities.

- A hospital appointment system stores information about patients, doctors, and appointments. A patient can book many appointments, but each appointment is booked by only one patient. A doctor can have many appointments, but each appointment is assigned to only one doctor. Each doctor belongs to one department, and one department can have many doctors.



Draw an Entity Relationship Diagram (ERD) according to the following scenario:

An online course platform stores information about students, courses, instructors, and enrollments. Each student has a student ID, name, email, phone number, and department. Each course has a course ID, course title, credit hours, and course category. Each instructor has an instructor ID, name, email, and designation. A student can enroll in many courses, and the same course can have many students. Each enrollment stores an enrollment ID, enrollment date, completion status, and final grade. Each course is taught by one instructor, but one instructor can teach many courses.